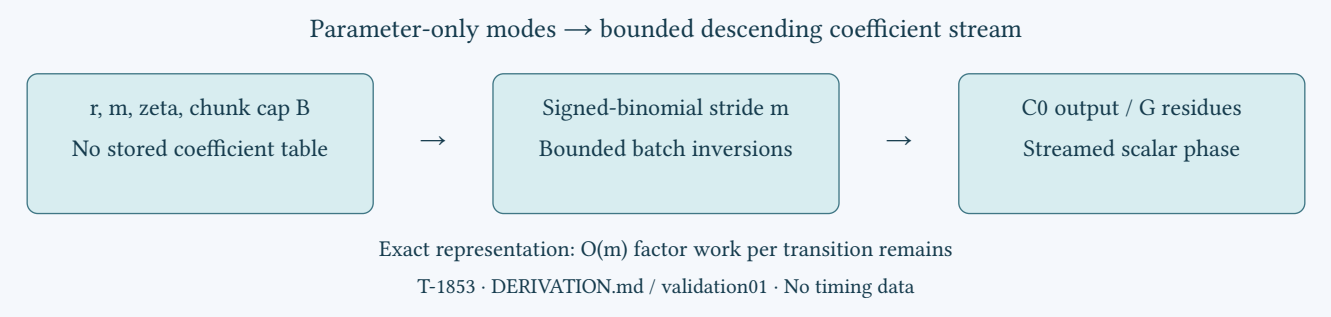
Stream mode coefficients into recovery

T-1853 · 2026-11-07 · Parameter-only prime-field mode representation

StreamModes stores r, m, zeta and a chunk cap. It generates selected signed-binomial coefficients in descending order, instead of compiling their full vectors. The first constraint is filled directly; later modes feed retained quotient-ring arithmetic, and phase extraction uses streamed scalar Horner. Public source and traced control flow are preserved.



Recurrence. For h=(r−1)/2 and c_N=(−1)^(h−N) binom(h,N), descending one selected coefficient multiplies by (−1)^m times the product of N−i, i=0..m−1, divided by the product of h−N+i, i=1..m. All factors lie in 1..h and are nonzero modulo prime r. The highest coefficient starts from the complementary binomial index q smaller than m. Bounded prefix/backward inversion supplies the ratios, retaining only the next coefficient between chunks.

Integration proof. The recurrence equals the original coefficient definition. Chunk Horner then produces exactly F_e(tZ) mod G. Quotient-ring projection and gcd invariance preserve both constraints; scalar Horner preserves phase evaluation. Unit and constant-mode cases retain their original meaning. DERIVATION.md gives the invariant and formulas.

Native check	Count	Result
Coefficient sequences	476	5,632 equal coefficients
Scaled residue comparisons	4,176	Exact equality
Scalar evaluations	1,392	Includes zero argument
Sequential state comparisons	42,324	Same state after each radius
Streamed public recovery	617	5,929 calls; zero unresolved

Storage and arithmetic. The native 64-bit mode object is 32 bytes. First C0 remains dense; later coefficient buffers are bounded by the chunk cap (or retained degree if smaller). Each coefficient transition uses m numerator and m denominator factors plus batch inversion/update work. Full mode generation has O(r) direct factor work and repeats on use. Degree-one residues may require an inversion per transition. This is a storage representation improvement with an explicit arithmetic tradeoff, not a timing result.

Evidence. validation01 receipt pins 45 transitive inputs and stdout; overflow checks enabled. Inherited controls pass. No native or report failure. PCC-20261107-d8b908 is an unsigned producer statement. No cloud run, timing measurement or full Cargo rerun; original 4,563-case cohort and trace protocol unchanged. The large r=2^61−1 check constructs only the parameter object, without coefficient generation or public recovery.

Next construction and sources. Replace direct stride-factor products with a faster bounded-storage recurrence evaluator. Exact predecessor source, NIST DLMF26.3 factorial/symmetry identities and FLINT modular-polynomial documentation were reviewed; KB unavailable. These are standard algebraic building blocks, with no novelty claim. Prior and canonical graphs remain unchanged.